

SURV 621/SURVMETH 621: WEB SURVEYS

November 4, 2024

Overview

- TSE Perspective on Web Surveys
- Basic Building Blocks of Web Survey Questionnaires
- Assignment #3 + Qualtrics Demo

TSE Perspective on Web Surveys

Coverage Error in Web Surveys

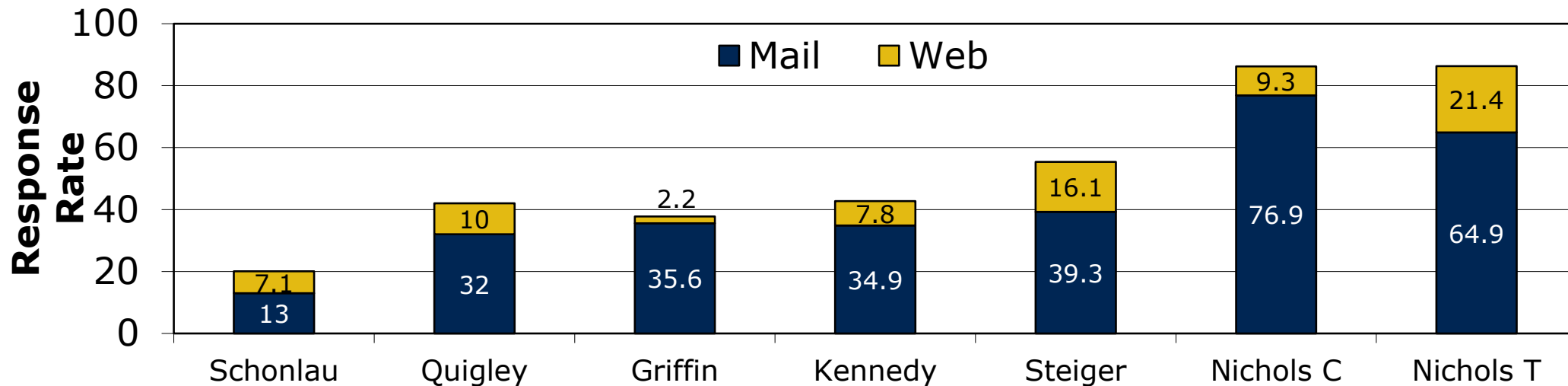
- Undercoverage largest concern in Web surveys of general population:
 - Persons without Internet access are, by definition, excluded from sampling frames
- Proportion of US adults who are offline has decreased steadily (Pew)
 - 48% in 2000, 5% in 2023
- Groups who were most likely to be offline in 2000 are still offline, although proportions have declined (Pew)
 - Older: 86% in 2000, 12% in 2023
 - Less educated: 81% in 2000, 10% in 2023

Nonresponse Error in Web Surveys

- Nonresponse error is function of nonresponse rate and difference between means for respondents and nonrespondents
- Cannot calculate response rate in opt-in samples
- Instead, participation or completion rate is recommended (Callegaro & Disogra, 2008): % starters who completed
- Response rate can be calculated in probability panels (but can be calculated in different ways)
- Focus typically on response rates in probability samples not nonresponse error

Response Rates in Probability Web Surveys

- Lozar Manfreda et al. (2008) meta-analysis of 24 studies shows that Web surveys yield response rate 11% lower than other modes (Phone, Mail, Fax, Email, IVR, TDE)
- Meta-analysis by Shih & Fan (2008) with 39 studies shows on average RR for mail 10% higher than list-based Web



Measurement Error in Web Surveys

- Web implementation of questionnaires may differ from other modes; web surveys are:
 - Self-administered (compared to In-Person and Phone interviews)
 - Computerized/automated and interactive (in contrast to Mail)

Web Surveys are Self-administered

- Advantages

- Reduction of social desirability and other biases associated with presence or behavior of *Iwer*
- Can be completed at *R*'s convenience

- Disadvantages

- No one is present to probe, clarify, assist, motivate, etc.
- Less guarantee that *R* will complete survey than in interviews
- Computing environment (context) not under researcher's control
- Web survey *Rs* at risk for distraction by multitasking (online and offline)

Web Surveys are Computerized and Interactive

- Complex routing, skips, edits, etc. possible
- Ability to randomize question and response option order
- Ability to tailor questionnaire for R by filling in name and previous answers
- Instrument can provide feedback (e.g., error messages, progress indicators)
- Can simulate some features of interviewer behavior (e.g., offering definitions, tailored encouragement, feedback to discourage speeding)
- Can provide include features that help respondent perform task (e.g., definitions on demand)

Basic Building Blocks of Web Survey Questionnaires

User-Centered Design Perspective

- The primary focus of the design process should be on the users (respondents) and their needs
 - Design process is iterative, often involving an evaluation of the product (web survey questionnaire) with potential users
- Design should facilitate and support the users in their task of completing the survey

Goal of Good Design

- The main tasks of the respondent are to read a question, respond to it, and navigate through the instrument
- Apply design features that support these tasks
- Main considerations: efficiency, accuracy, and satisfaction

Range of available software for developing computerized survey instruments

- *Blaise*
 - Supports CATI, CAPI, CASI and web surveys
 - Developed by Statistics Netherlands; used by many national statistical offices, including U.S. Census Bureau
- *Qualtrics*
 - Online survey platform
 - Users can create surveys for free (features are restricted) or buy license to use all features
 - Customers include universities and private companies (market research and other applications)
- *LimeSurvey*
 - Open-source software
 - Two software formats: online platform and software for download
 - Users can create surveys for free (restricted to 25 responses/month) or enroll in paid plans
- *Survey Monkey*
 - Online survey platform
 - Users can create online surveys for free; company makes money by selling back-end services

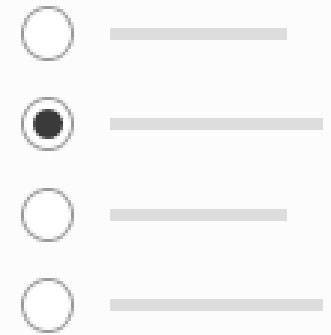
Basic Question Types

- Radio buttons and checkboxes
- Drop-boxes (drop-down list)
- Text fields
- Sliders scales
- Grids

Radio Buttons and Check Boxes

- Radio buttons
 - Standard format for single-choice responses
 - When one item in a list is selected, all others are unselected
- Check boxes
 - Appropriate for multiple response, check-all-that apply questions
 - Can toggle several items in list on or off

RADIO BUTTONS



Four radio button options are shown, each with a horizontal line to its right. The second option from the top is selected, indicated by a solid black dot in the center of the circle.

CHECKBOXES



Four checkbox options are shown, each with a horizontal line to its right. The second and fourth options from the top are checked, indicated by a checkmark inside the square box.

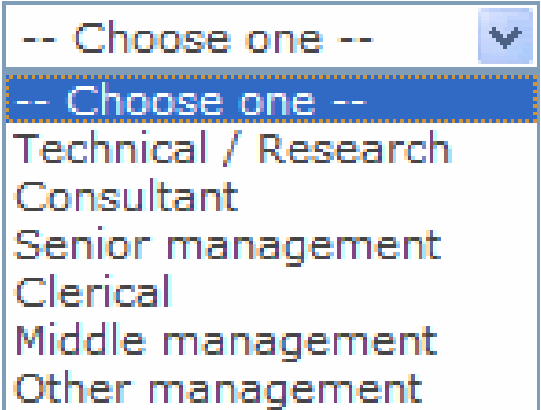
Radio Buttons and Check Boxes Cont'd

- Advantage: used in many applications and so familiar to most users
- Design considerations
 - Use the appropriate tool for the task
 - More appropriate for shorter lists than longer lists
 - Presents a small target which can be difficult to select, especially on smartphones

Drop-Boxes

- Used for single-choice responses
- When box is inactive, it displays a single value. When activated, it displays a list of values.
- Advantage: saves space on screen

What is your job function:



The image shows a web form with a label "What is your job function:" followed by a dropdown menu. The dropdown menu is currently open, displaying a list of job functions. The first option, "-- Choose one --", is highlighted in blue. The other options are "Technical / Research", "Consultant", "Senior management", "Clerical", "Middle management", and "Other management".

Job Function
-- Choose one --
Technical / Research
Consultant
Senior management
Clerical
Middle management
Other management

Drop-Boxes Cont'd

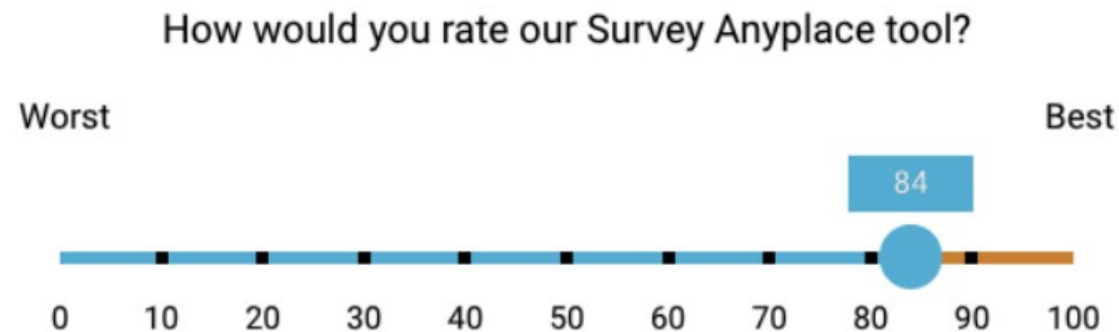
- Appropriate when:
 - The list is too long to display on the page
 - The answer is known to the respondent
 - The list is ordered in a meaningful way (e.g., alphabetic)
 - Selecting the response is more efficient than typing
 - The answer is still recognizable when truncated

Text Fields

- Two formats:
 - Single-line text field for numbers or short text answers
 - Text area for longer, open responses
- Designer can control:
 - Size of visible area (how big the box looks)
 - Maximum input (how much can be entered)
- Design considerations
 - May need to provide guidance on format of information to be entered (e.g., MM/DD/YYYY)
 - Provide sufficient space for input but not too much space

Slider Scales

- Used for continuous scales (e.g., 1-100)
- Not appropriate for ordinal response scales (e.g., Likert scale)
 - Evidence of high missing data and longer completion times (Funke 2016)
- Design considerations
 - Answers might be influenced by initial position of slider (Funke 2011)
 - Answers might be influenced by feedback about selected value (Liu & Conrad, 2016)
 - Slider line is relatively short when displayed on smartphones



Grids or Matrix Question

- Table displaying set of items sharing the same response options
- Items are presented in rows and response options in columns
- Respondents answer row-by-row

	Please rate the following aspects of the motivational seminar that you attended		
	Highly motivating	Motivating	Not motivating
Guest speaker	☐	☐	☐
Seminar pamphlets	☐	☐	☐
Motivational CD	☐	☐	☐

www.zarca.com

Grids Cont'd

Advantages

- **Faster to complete** (Couper et al. 2001; Mavletova & Couper 2015; Rossmann et al. 2018; Toepoel & Dillman 2008; Toepoel et al. 2009; Tourangeau et al. 2004, 2013)
 - Reduce screen orientation time
- **Provides more context for related items**
 - Higher inter-item correlation when multiple items measure same construct (Tourangeau et al. 2004)

Disadvantages

- **Less *R* satisfaction** (Thorndike et al. 2009; Toepoel et al. 2009)
 - Especially for large grids (Couper et al. 2013; Liu & Cernat 2016)
- **May encourage straightlining and non-differentiation** (Baker et al. 2010; Couper et al. 2013; Rossmann et al. 2018; Zhang & Conrad 2014)
- **May increase item missing data** (Mavletova & Couper 2015; Rossmann et al. 2018; Peytchev et al. 2006; Toepoel et al. 2009)
- **May not fit screen width of smartphone**

Avoid Large Grids

Wie zufrieden sind Sie mit den folgenden Flug- und Bahngesellschaften in Bezug auf die folgenden Kriterien?
Bitte bewerten Sie dies auf einer Skala von "--" (sehr unzufrieden) bis "++" (sehr zufrieden).

	 Austrian Airlines					 ÖBB				
	--	-	0	+	++	--	-	0	+	++
Anzahl meiner Reiset Teilnehmer	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Komfort des Reisens und Wartens	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Sicherheit des Verkehrsmittels	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Kosten (günstiger Preis)	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Schnelligkeit/Zeitdauer	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Beschäftigungsmöglichkeiten während der Reise	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Art und Menge des Gepäcks (Mitnahmemöglichkeit)	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Möglichkeit zu Spontaneität und Flexibilität	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Reisen mit Kunden	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Ökologische Verantwortung	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Einfachheit und Stressfreiheit	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Reise ins Ausland	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

	 Austrian Airlines					 ÖBB				
	--	-	0	+	++	--	-	0	+	++
Betreuung und Unterstützung (während der Reise)	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Selbstbestimmtheit und Unabhängigkeit	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Direktheit (ohne Umsteigen)	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Planbarkeit der Reise und Pünktlichkeit	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Ruhe und Ungestörtheit	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Passende Abfahrts- / Ankunftszeit	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Page layout and design

- Grouping of items
- Navigation
- Designing for mobile devices

Grouping of Items: Scrolling vs. Paging

- Scrolling Web survey
 - Survey questions are presented on one single page
 - *R* can scroll back and forth through instrument
- Paging Web survey
 - Survey questions presented on separate pages
- Alternative: grouping thematically related questions on same page
 - e.g., multi-item scale, sociodemographic Qs

Strength of Each Approach

Scrolling (static)

- Easy to program
 - Minimizes page loading time
 - *R* can review entire questionnaire
 - Aware of progress / context
 - Can easily review/change previous answers
- Well suited for shorter, simpler Web surveys

Paging (interactive)

- Complex skips, fills, and other logic possible
 - Order of answering controlled
 - Less item missing data (Lozar Manfreda et al. 2002; Peytchev et al. 2006)
 - All data captured in progress – can be suspended and resumed
- Well suited for complex Web surveys

Navigation

- Four actions generally possible in paging surveys:
 - “Next”: Proceed to next page
 - “Back”: Return to previous page
 - Quit, stop, or suspend
 - Launch separate form or page with procedural help, definitions, etc.
- Design tips
 - Clearly label action buttons and place them in a logical and consistent position
 - Permit respondents to back up and review/change answers, but don’t encourage this behavior
 - No need for question numbering (if skips are automated)

Assignment #3 + Qualtrics Demo

Assignment #3

Collaborate with team members to:

1. Program preliminary version of instrument
2. Briefly describe each component of your testing approach:
 - question-by-question testing
 - scenario testing
 - testing on different browsers/devices
3. Test and record comments (in spreadsheet)
4. Revise instrument accordingly

Qualtrics: Getting started

- University of Maryland faculty, staff and students have access to Qualtrics at: <https://umdsurvey.umd.edu>
- University of Michigan faculty, staff and students have access to Qualtrics at: <https://umich.qualtrics.com>
- Lots of online help available on Qualtrics website

Qualtrics Demo: Creating Q'naire; Importing & Exporting QSFs

- New questionnaire
 - “Create new project” -> “Survey” -> “Get Started” -> create a project name
 - Or to import an existing q'naire: select “Import a QSF file” -> select QSF
- Sharing a questionnaire with group members
 - “Tools” -> “Collaborate” -> enter email address
- Export
 - “Tools -> “Import/Export” -> “Export Survey”
- Generating a survey link
 - “Distributions” -> “Anonymous Survey Link”

Qualtrics Demo Cont'd: Some options for controlling question wording, flow and presentation

- Survey options -> Responses
 - Back button*
- Survey look and feel
 - Progress bar*
- Question options
 - Response requirements
 - Page break
 - Skip logic/display logic*
 - Piped text
- Block options
 - *Loop and merge* used to dynamically repeat a block of questions multiple times for a respondent
 - *Loop and merge fields* used to create dynamic text that is different for each loop
 - > `${Im://Field/1}` `${Im://Field/2}` ...
 - Display logic within loop and merge
 - Randomizer

*Free version of software provides no access to these features.

Download the survey instrument from Canvas (assignment #3)

What question types are used?

Does it use a paging or scrolling design?

Tips for Using Blocks in Assignment #3

- Put first three questions in one block
 - Block 1: Consent; Q1; Q2
- Create three separate blocks with *loop and merge* based on Q2*
 - Block 2: Q3_X (relationships)**
 - Block 3: Q4_1 through Q8_1 (demographics)
 - Block 4: Q10_1 through Q16_1 (Employment)
- Create one block with *randomizer*
 - Block 5: Q17A and Q17B

*This will require creating a table containing loop and merge text fields. See: <https://rb.gy/ax2lpc>

**Use display logic to hide this question during the first loop

-You may find that the skips are easier to program by adding *display logic* rather than *skip logic*

-Consent can be implemented by using a “multiple choice” question type (allow multiple answers), only displaying 1 response choice, adding a “force choice” requirement

-Some numeric entry question require “validation” (e.g., to restrict values from 0-100)

Instrument testing protocol

- Question-by-question testing: Each question or screen in the survey should be tested before moving on to the next.
 - Looking primarily for issues with response selection and text and text fills
- Scenario testing to evaluate display logic and skips: Construct scenarios of responses, then enter accordingly
 - Construct scenarios that test all of the different text fills and skips
 - Include missing values and out-of-range values
- Instrument should be tested on different machines (desktops or laptops, tablets and phones) using different browsers (Chrome, Firefox and Internet Explorer)
 - Looking primarily for formatting issues

Tips for Scenario Testing in Assignment #3

- Key factors for fills
 - whether question is asking about respondent themselves or someone else in the household
 - whether question is about someone's main job or secondary job
- Key factors for scenarios
 - household size
 - working or not
 - job identified based on reporting to work last week (Q10) or identified based on having a job from which temporarily absent (Q11)
 - multiple jobs or not
 - whether year of birth is recorded
- Example scenarios:
 - respondent is the only member of their household and has two jobs
 - two people in household, one with job identified based on reporting work last week (Q10) and one with job identified based on having a job from which temporarily absent (Q11)